

4.14.1 Overview

4.14.1.1 Purpose of the project

The project allows students to develop their practical skills in the context of solving a realistic problem or carrying out an investigation. The project is intended to be as much a learning experience as a method of assessment; students have the opportunity to work independently on a problem of interest over an extended period, during which they can extend their programming skills and deepen their understanding of computer science.

The most important skill that should be assessed through the project is a student's ability to create a programmed solution to a problem or investigation. This is recognised by allocating 42 of the 75 available marks to the technical solution and a lower proportion of marks for supporting documentation to reflect the expectation that reporting of the problem, its analysis, the design of a solution or plan of an investigation and testing and evaluation will be concise.

4.14.1.2 Types of problem/investigation

Students are encouraged to choose a problem to solve or investigate that will interest them and that relates to a field that they have some knowledge of. There are no restrictions on the types of problem/investigation that can be submitted or the development tools (for example programming

language) that can be used. The two key questions to ask when selecting a problem/investigation are:

- Does the student have existing knowledge of the field, or are they in a position to find out about it?
- Is a solution to the problem/investigation likely to give the student the opportunity to demonstrate the necessary degree of technical skill to achieve a mark that reflects their potential?

Some examples of the types of problem to solve or investigate are:

- a simulation for example, of a business or scientific nature, or an investigation of a well-known problem such as the game of life
- a solution to a data processing problem for an organisation, such as membership systems
- the solution of an optimisation problem, such as production of a rota, shortest-path problems or route finding
- a computer game
- an application of artificial intelligence
- a control system, operated using a device such as an Arduino board
- a website with dynamic content, driven by a database back-end
- an app for a mobile phone or tablet
- an investigation into an area of computing, such as rendering a three-dimensional world on screen
- investigating an area of data science using, for example, Twitter feed data or online public data sets
- investigating machine learning algorithms.

There is an expectation that within a centre, the problems chosen by students to solve or investigate will be sufficiently different to avoid the work of one student informing the work of another because they are working on the same problem or investigation. Teachers will be required to record on the Candidate Record Form for each student that they have followed this guideline. If in any doubt on whether problems chosen by students have the potential to raise this issue, please contact your AQA adviser.

Table 1 and Table 2 show the technical skills and coding styles required for an A-level standard project. If a problem/investigation is selected that is not of A-level standard then the marks available in each section will be restricted.

4.14.1.3 Project documentation structure

The project is assessed in five sections. The table below lists the maximum available mark for each section of the project:

Section		Max mark
1	Analysis	9
2	Documented design	12
3	Technical solution	42
4	Testing	8
5	Evaluation	4
Total		75

For marking purposes, the project documentation should be presented in the order indicated in the table above. The table does not imply that students are expected to follow a traditional systems life cycle approach when working on their projects, whereby a preceding stage must be completed before the next can be tackled. It is recognised that this approach is unsuited to the vast majority of project work, and that project development is likely to be an iterative process, with earlier parts of the project being revisited as a result of discoveries made in later parts. Students should be encouraged to start prototyping and writing code early on in the project process. A recommended strategy is to tackle the critical path early in the project development process. The critical path is the part of the project that everything else depends on for a working system or a complete investigation result to be achieved.